

虛擬世界之設計與研究 Virtual World Design and Research

A-Frame WebVR Tutorial:
Adding 3D Models

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Overview

- Other than primitive shapes, you can also add 3D models in an A-Frame scene
- The overall workflow is:
 1. Prepare the 3D models
 2. Set up you're A-Frame HTML file so it has a "server"
 3. Add codes in your HTML
 4. Run/test the HTML file

1. Prepare the 3D Models

- Sources of 3D models

- Created from your own 3D modeling software (e.g., Blender, Maya)
- Or, download free/paid 3D models from websites like [SketchFab](#), [TurboSquid](#), [CGTrader](#), etc.

- Preferred filetype

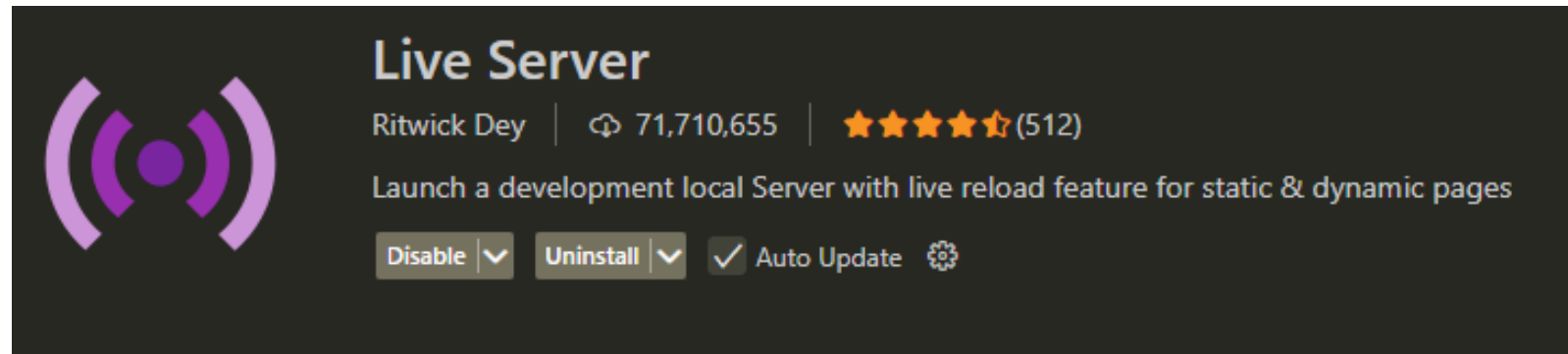
- A-Frame works best with .glTF or .glb files
- It can also work with .obj files, but you need to get the obj file's accompanying .mtl file. (Even with a 3D model's obj and mtl files, it may not always display correctly because sometimes you need the texture file specified, so it's best to just use glTF or glb)

2. Set up you're A-Frame HTML file so it has a “server”

- A server is needed, because by default, browsers cannot directly show the 3D models you have added with the HTML file
- External server
 - Glitch: Its A-Frame hosting service ended in July 2025....
 - GitHub
 - A-Frame's official website's pages: [3D Models](#), [Hosting & Publishing](#)
- Local server
 - In this tutorial, we will focus on using this method

2. Set up you're A-Frame HTML file so it has a “server”

- Assuming you are using VS Code:
 - Click the "Extensions" box on the left sidebar.
 - Search for "Live Server" (by Ritwick Dey) and install it.
 - Feel free to try other extensions

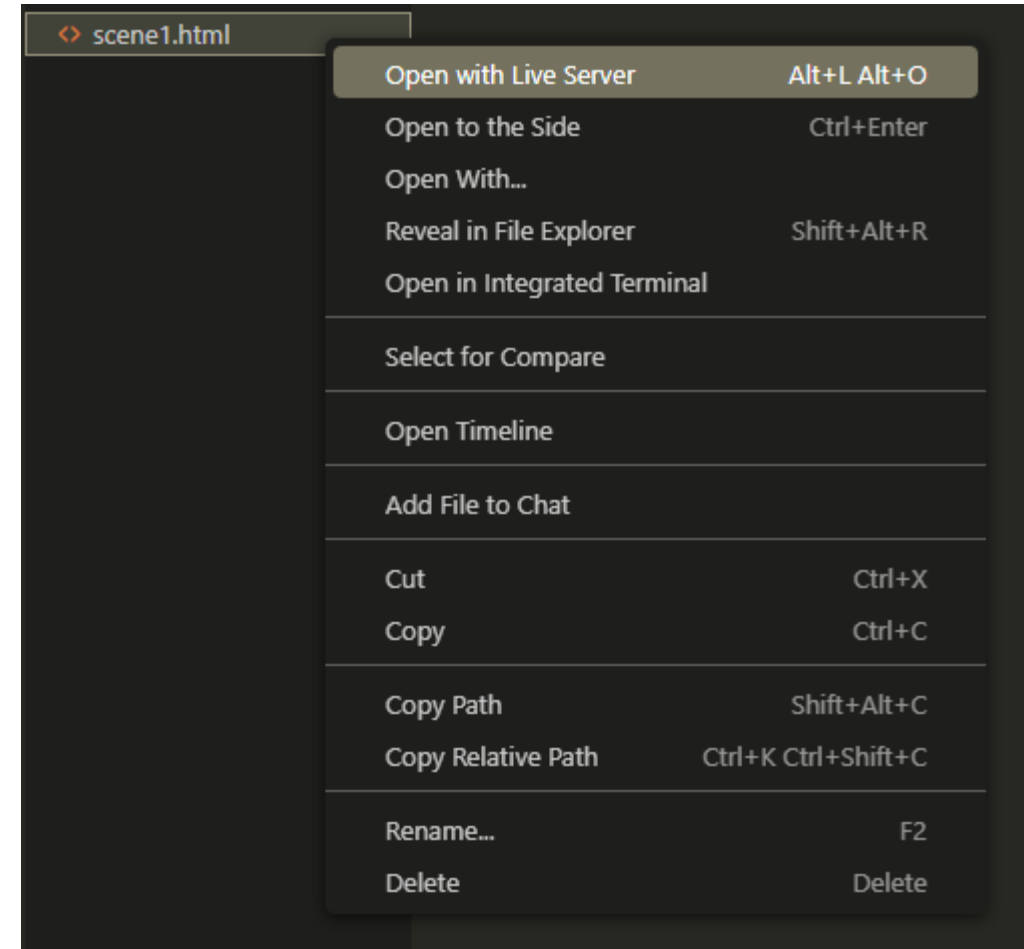


3. Add codes in your HTML

- Add codes in your HTML to specify how and what you are using for your 3D models
- Remember to organize your 3D models, preferably they are in the same sub-folder
- Please see the example HTML file and its commenting accompanied with this tutorial

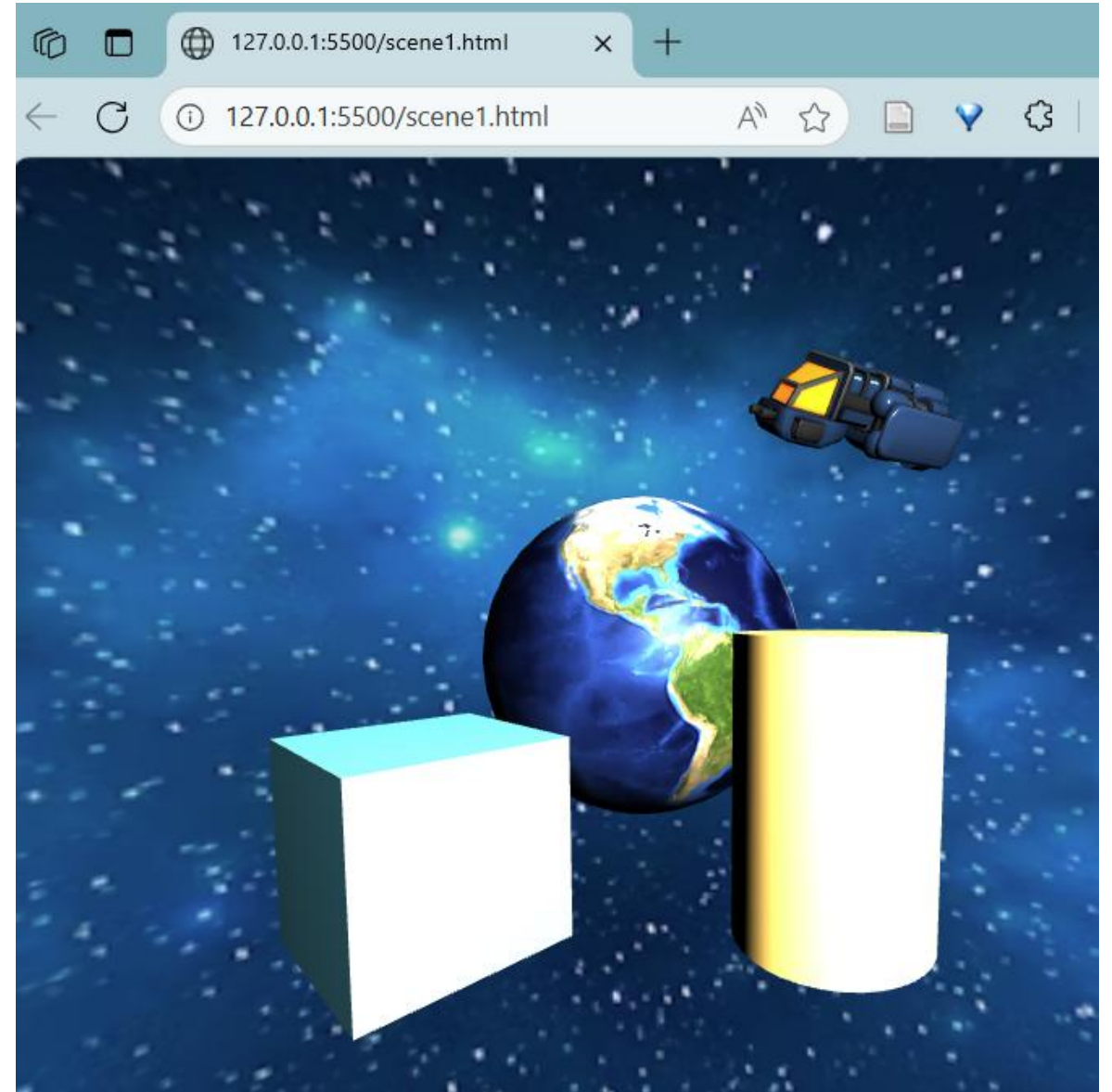
4. Run/test the HTML file

- Assuming you are using VS Code
 - Whenever you want to test your modified HTML file, right-click it and choose "Open with Live Server"
 - A browser window (or tab) will pop up to show the result of latest HTML file



Result

- When the HTML runs, it uses a local server, and displays the spaceship 3D model



Thank You!
Questions?

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